

No. of Functional Features Included in the Solution

Date	5 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration	Allows new users to create an account secu	Authentication	Done	High
2	User Login	Authenticates registered users to access the system.	Authentication	Done	High
3	Crop Prediction	Predicts the most suitable crop based on soil and environmental parameters using a Machine Learning model.	Prediction Module	Done	High
4	Crop Recommendation Result	Displays the predicted crop along with recommendation details.	Results Module	Done	Medium
5	User Dashboard	Provides access to prediction history and user information.	Dashboard	In Progress	Medium
6	Responsive User Interface	Ensures the application works on desktop and mobile devices	Frontend	Done	Medium
7	Prediction History	Stores and displays previous crop predictions.	Database	In Progress	Medium
8	Weather API Integration	Fetches weather information to improve recommendations	External API	Pending	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8

Total Features Implemented	5
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	5

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Registration, Responsive Dashboard
2	Backend / Logic	2	Login Authentication, Crop Prediction
3	Database / Storage	2	User Data, Prediction History
4	API / Integration	1	Weather API Integration
5	Security / Authentication	1	User Login & Registration